



St Teresa's School
- Founded by the Sisters of Mercy in 1930 -



Tour de Maths 2024

Leg 2

29 February 2024

MEMO



Sponsored by:

CASIO

5 MARK QUESTIONS

QUESTION 1

What is the smallest positive integer that is a multiple of four, six and ten?

By Prime factorisation

$$4 = 2^2$$

$$6 = 2 \times 3$$

$$10 = 2 \times 5$$

$$\begin{aligned} LCM &= 2^2 \times 3 \times 5 \\ &= 60 \end{aligned}$$

QUESTION 2

$2021 - 2020 + 2019 - 2018 + \dots + 5 - 4 + 3 - 2 + 1$ equals

By Grouping

$$\begin{aligned} &(2021 - 2020) + (2019 - 2018) + \dots + (5 - 4) + (3 - 2) + 1 \\ &= 1 \end{aligned}$$

QUESTION 3

The time 2021 minutes after 20:21 on a digital 24-hour clock is

$$2021 \text{ minutes} \div 60 \text{ min/h}$$

$$\approx 3 \text{ hours}$$

$$\text{The remainder minutes} = 2021 - (33 \text{ hrs} \times 60 \text{ minutes/h})$$

$$= 1$$

$\therefore$ After 33 hrs, we have 1 minute remaining

Lets add the hours and minutes to 2021

$$20 \text{ hrs} + 33 \text{ hrs} = 53 \text{ hrs}$$

$$21 \text{ min} + 1 \text{ min} = 22 \text{ min}$$

$$53 \text{ hrs} + 10 \text{ hrs} = 63 \text{ hrs}$$

$\therefore$ So 2021 minutes after 20:21 on a digital 24-hour is 06:02

QUESTION 4

If January 1st 2019 was a Tuesday, how many Tuesdays were there in 2019?

The sequence 1, 8, 15, 22, 29, 36, ... is an arithmetic sequence with:

$$a = 1 \text{ and } d = 7.$$

The number of Tuesdays n is the largest value of n

for which the n th term is less than or equal to 365.

$$a + (n - 1)d \leq 365$$

$$1 + (n - 1)7 \leq 365$$

$$n \leq 53$$

$\therefore$ There are 53 Tuesdays.

QUESTION 5

The positive integers are written in a long sequence 12345678910111213 ... When the sequence contains 100 digits, how many of them are 1's?

Group from 1 to 9, 10 to 19, 20 to 29, and so on.

The contributions of these groups to the number of digits are 9, 20, 20, 20, 20, 20, ...

so the 100th digit is in the 6th group, that is, from 50 to 59. The last 1 is at 51.

The number of 1's altogether is therefore: $1 + 11 + 1 + 1 + 1 + 1 = 16$.

(All groups have one 1, except the second, which has one as a units digit & ten as tens digits.)

QUESTION 6

Your sister will be x years old in the year x^2 . In what year was your sister born?

Try the first year from now that is a square. It is $45^2 = 2025$.

So your sister was born in the year $2025 - 45 = 1980$.

(The next square is 2116, which will give the year of birth as $2116 - 46 = 2070$, which is impossible.)

QUESTION 7

After how many hours will the second candle be twice the height of the first candle, given that both were lit simultaneously and one burns out in 3 hours while the other takes 4 hours to burn completely?

Let the height of each candle in the beginning be x units.

The first candle burns down x units in 3 hours, so $\frac{x}{3}$ units in one hour.

So it burns down $\frac{xt}{3}$ at units in t hours, and its height after t hours is $x - \frac{xt}{3}$.

Similarly the height of the second candle is $x - \frac{xt}{4}$ after t hours.

For the second candle to be twice the height of the first,

we must have $x - \frac{xt}{4} = 2 \left(x - \frac{xt}{3} \right)$

Solving for t , we get $t = 2,4 \text{ hours} = 144 \text{ minutes}$.

QUESTION 8

What is the tens digit of $N = 1 \times 3 \times 5 \times 7 \times \cdots \times 99$?

The number is odd and also an odd multiple of 25, so ends in 25 or 75.

(The even multiples of 25 end in 50 or 00).

If it is of the first type, then it is congruent to 1 mod 4, and if it is of the second type, then it is congruent to 3 mod 4, which is also $(-1) \bmod 4$. So let reduce the number mod 4.

$$N = 1 \times 3 \times 5 \times 7 \times \cdots \times 99$$

$$= (1 \times 5 \times 9 \times \cdots \times 97) \times (3 \times 7 \times \cdots \times 99)$$

$$\equiv_4 1^{25} (-1)^{25} \equiv_4 (-1) \equiv_4 3.$$

so $N = 4k + 3$ cannot end in 25.

Answer is 75.

QUESTION 9

Determine the number of possible distinct pairs of integers $(a; b)$ for which

$$\left(2a + 5b - \frac{1}{2}\right)^2 \leq 2 - (a + 2b)^2$$

Let $u = a + 2b$ and $v = 2a + 5b$

But since a, b are integers then

$$(u; v) = \{(-1; 0), (-1; 1), (0; 0), (0; 1), (1; 0), (1; 1)\}$$

Solving for a and b in terms of u and v , we get $a = 5u - 2v$, $b = v - 2u$

Substituting for u, v , we obtain 6 solutions for $(a; b)$, which are:

$(a; b)$, namely $(-5; 2), (-7; 3), (0; 0), (-2; 1), (5; -2)$ and $(5; -1)$

QUESTION 10

Find the value of $p + q$ if $p, q \in \mathbb{Z}^+$ and $p + q + pq = 54$.

$$p + q + pq = 54$$

$$p(1 + q) = 54 - q$$

$$p = \frac{54 - q}{1 + q}$$
$$= \frac{55 - (1 + q)}{1 + q}$$

$$\frac{55}{1 + q} - 1 \text{ is an integer}$$

So $1 + q$ is a factor of 55 and moreover, $1 + q > 0$ and $1 + q < 55$.

Since both p and q are positive integers, then $1 + q = 5$ or 11

then p is either 10 or 4. So $p + q = 10$ or $4 + 10 = 14$

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QUESTION 11

John, Simon and Peter decide to paint a wall. John could have painted it himself in 180 minutes, while Simon would have taken 4 hours and working together John, Simon and Peter can paint it in 1 hour 20 minutes. How long would it take the Peter to paint the wall alone?

In an hour John can paint a third of the wall, Simon can paint a quarter of the wall while Peter can paint a sixth of the wall. So together they can pain

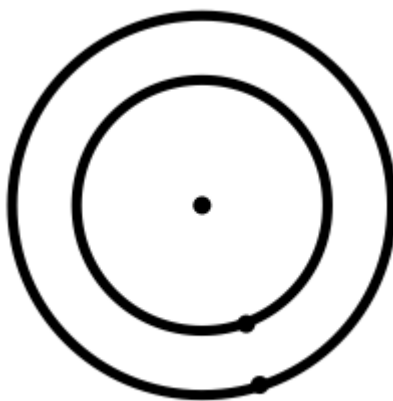
$$\frac{1}{3} + \frac{1}{4} + \frac{1}{6} = \frac{3}{4} \text{ of the wall in an hour}$$

Meaning that they can paint a quarter of the wall in $\frac{60}{3} = 20$ minutes.

*So to paint the whole wall they would need $20 \times 4 = 80$ minutes
= 1 hour and 20 minutes*

QUESTION 12

The diagram shows two concentric circles. If the circumference of one exceeds the other by 6 cm by how much does its radius exceed the other radius?



Let r be the radius of the small circle, and R the radius of the large circle.

Their circumferences are therefore $2\pi r$ & $2\pi R$. The circumferences differ by 6,

$$\text{So } 2\pi(R - r) = 6$$

$$R - r = \frac{6}{2\pi} = \frac{21}{22}$$

QUESTION 13

Determine the number of pairs $(a; b)$ of integer solutions for $2^{2a} = 55 + 3^{3b}$

$$2^{2a} - 3^{3b} = 55$$

$$= (2^a)^2 - (3^b)^2$$

$$= (2^a - 3^b)(2^a + 3^b) = 55$$

The first factor is the smaller than the second so we have

$$2^a - 3^b = 1 \text{ and } 2^a + 3^b = 55$$

OR

$$2^a - 3^b = 5 \text{ and } 2^a + 3^b = 11$$

By adding the first two we get $2^{a+1} = 56$ which is a power of 2.

From the second pair of equations, $2^{a+1} = 2^4$

$$a + 1 = 4$$

$$a = 3$$

$$3^b = 11 - 8 = 3$$

$$b = 1$$

$$(3; 1)$$

QUESTION 14

The function $\gamma(x)$ is defined for all $x \in \mathbb{R}$. If $\gamma(a + b) = \gamma(a)\gamma(b)$ for all a, b and $\gamma(1) = 2$.

Find the value of $\gamma(2013)$.

$$\begin{aligned}\gamma(x + 1) &= \gamma(x)\gamma(1) = 2\gamma(x) \text{ for any number } x \\ \gamma(2013) &= 2\gamma(2012) = 2^2\gamma(2011) = 2^3\gamma(2010) \dots \\ &= 2^m\gamma(2013 - m) = \dots = 2^{2012}\gamma(2013 - 2012) \\ &= 2^{2012}\gamma(1) \\ &= 2^{2013}\end{aligned}$$

QUESTION 15

Find the number of subsets of $X = \{1,2,3,4,5,6,7,8\}$ that are subsets of neither $Y = \{1,2,3,4,5\}$, nor $Z = \{4,5,6,7,8\}$

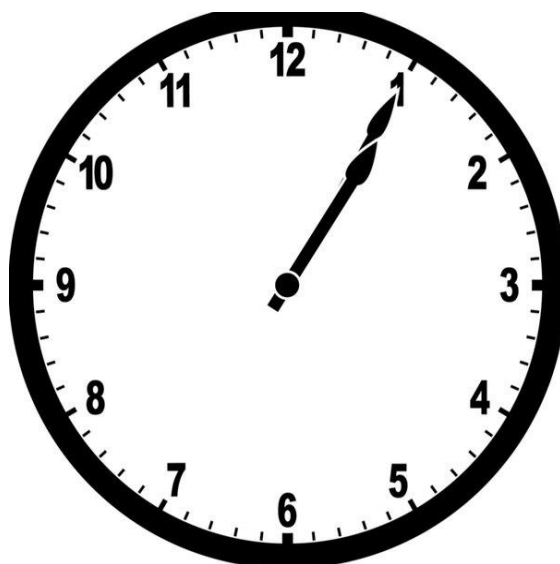
Consider the sets $\{1,2,3\}$ and $\{6,7,8\}$. We must choose at least elements from both of these which can do $2^3 - 1 = 7$ ways.

Now for the set $\{4,5\}$ we can choose any including the empty set subset so we have $2^2 = 4$ choices. So the total number of subsets is $7 \times 7 \times 4 = 196$

20 MARK QUESTIONS

QUESTION 16

Between 12:00 and 13:00, there are two times when the hands of a clock are exactly at right angles. How many minutes apart are these two times?



Observe that the hour hand moves through 5 minutes during the same time that the minute hand moves through 60 minutes. So the minute travels 12 times faster than the hour hand.

When the minute hand moves through x minutes, the hour hand covers $\frac{x}{12}$ min,

so the number of minutes separating them is $x - \frac{x}{12} = \frac{11x}{12}$,

but each minute is $\frac{1}{60}(360^\circ) = 6^\circ$, so the angle between them is $\left(\frac{11x}{12}\right) 6^\circ = \frac{11x}{2}^\circ$

When they are at right angles to each other we have either $\frac{11x}{2} = 90^\circ$ or $\frac{11x}{2} = 270^\circ$

So x is either $\frac{180}{11}$ or $\frac{540}{11}$ minutes. The number of the minutes between these two times is

$$\frac{360}{11} = 32\frac{8}{11} \text{ minutes}$$

QUESTION 17

Neither of the two integers x and y is divisible by 3. What are the possible remainders when $x^3 + y^3$ is divided by 9?

Let us look at x . On division by 9, the remainders can be none of 0, 3 and 6,

for otherwise m would be divisible by 3. So

$$x \equiv_9 1, 2, 4, 5, 7, 8$$

$$x^3 \equiv_9 1, 8, 64, 125, 343, 512, \equiv_9 1, 8, 1, 8, 1, 8$$

That is, the cube of m is congruent to either 1 or 8. The same applies to the cube of y .

$$\text{So } x^3 + y^3 \equiv_9 2, 9, 16 \equiv_9 0, 2, \text{ or } 7$$

The only possible remainders are 0; 2; or 7

QUESTION 18

From the numbers 1, 2, 3, ..., 500, a sequence is formed by deleting numbers so that no two of the remaining numbers have a sum which is a multiple of 7.

What is the maximum number of numbers in this sequence?

Let us try to construct such a sequence. Include 1, 2, 3 and 7. None of 4, 5 or 6 is now allowed.

We can include 8, 9 and 10, but not 11, 12, 13 or 14.

Next we have 15, 16, 17. Notice that mod 7, the sequence is 1, 2, 3, 0, 1, 2, 3, 1, 2, 3, ...

So from 1 to 497 (that is, 71 groups of 7), each group contributes three numbers,

save the first which includes the number 7. That makes $71(3) + 1 = 214$.

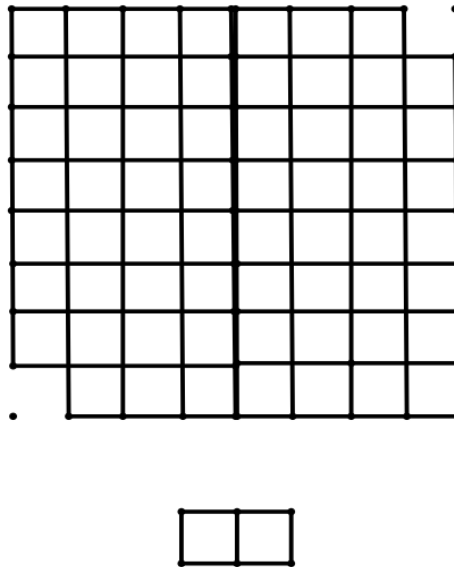
The numbers 498, 499 and 500, leave remainders 1, 2 and 3 so they too must be counted in.

So the longest sequence has 217 numbers.

QUESTION 19

Two opposite corners of an 8×8 grid are removed, so 62 squares are left.

You have thirty-one 1×2 dominos. Place the dominos on the grid in such a way that all 62 squares on the grid are occupied.



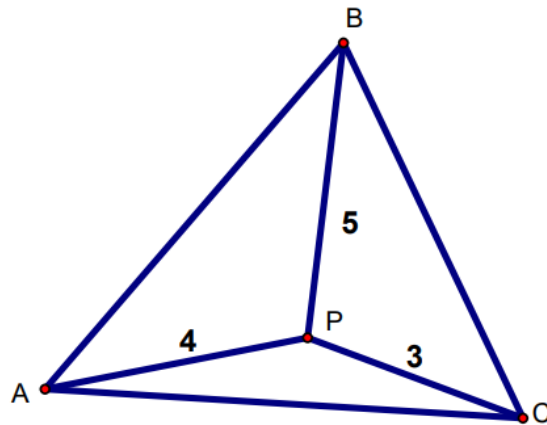
The trick is to paint the grid as if it were a chessboard, as in the diagram alongside.

Note that the removed squares are both black, so we have 30 black and 32 white squares. But the 31 dominos will cover 31 black and 31 white squares!

We conclude that it is impossible to place the dominos as required.

QUESTION 20

A point P has distances 3, 4 and 5 from the vertices of an equilateral triangle, as shown. Calculate the length of one side of the equilateral triangle.



Rotate $\triangle CBP$ through 60 degrees in an anticlockwise direction about C.

It becomes $\triangle CAF$. Let $AE \perp CP$, with E on CP extended.

In the rotation, B goes to A and P goes to F, say. Hence $AF = 5$ &

$PF = 3$, so by Pythagoras we have that $\angle APF = 90^\circ$.

Since CPF is a straight line, angle $APE = 180 - 60 - 90 = 30^\circ$

The other two sides of right angled triangle APE can now be easily seen to be

$AE = 2$ and $EP = 2\sqrt{3}$.

The two sides of right – angled triangle AEC are 2 and $3 + 2\sqrt{3}$

Hence its hypotenuse, which is the side AC of equilateral triangle ABC,

is $AC = \sqrt{25 + 12\sqrt{3}}$
